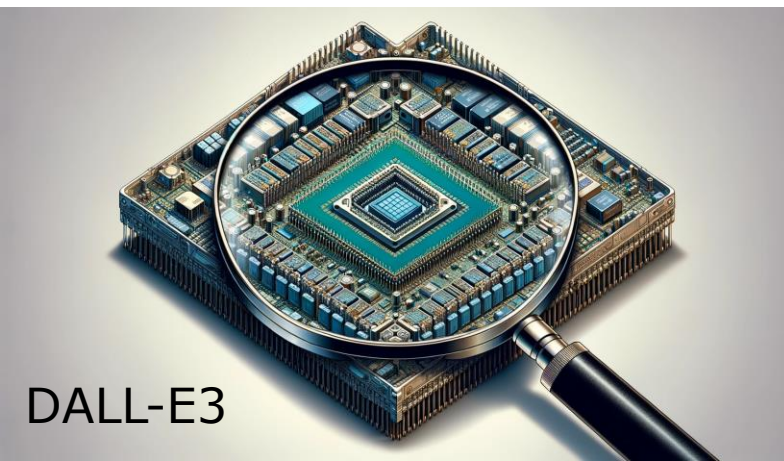
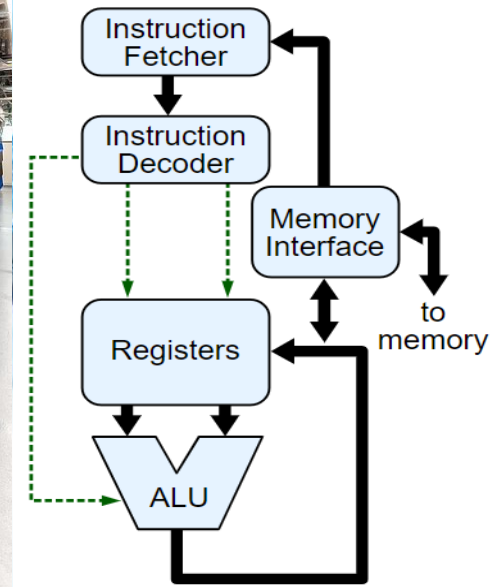
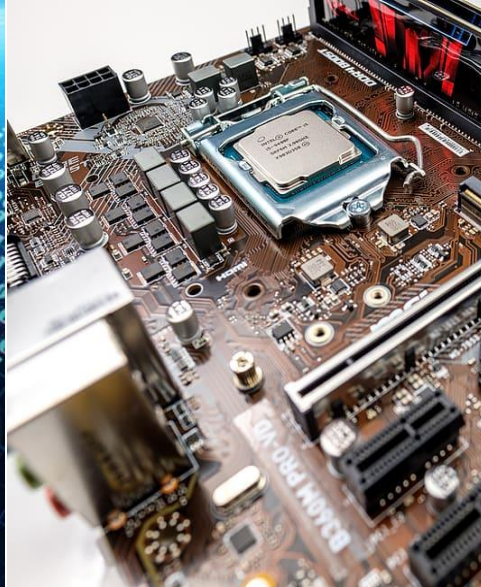




DALL-E3

Check In ?



Code
Class Code

Code
514006

Memory Units



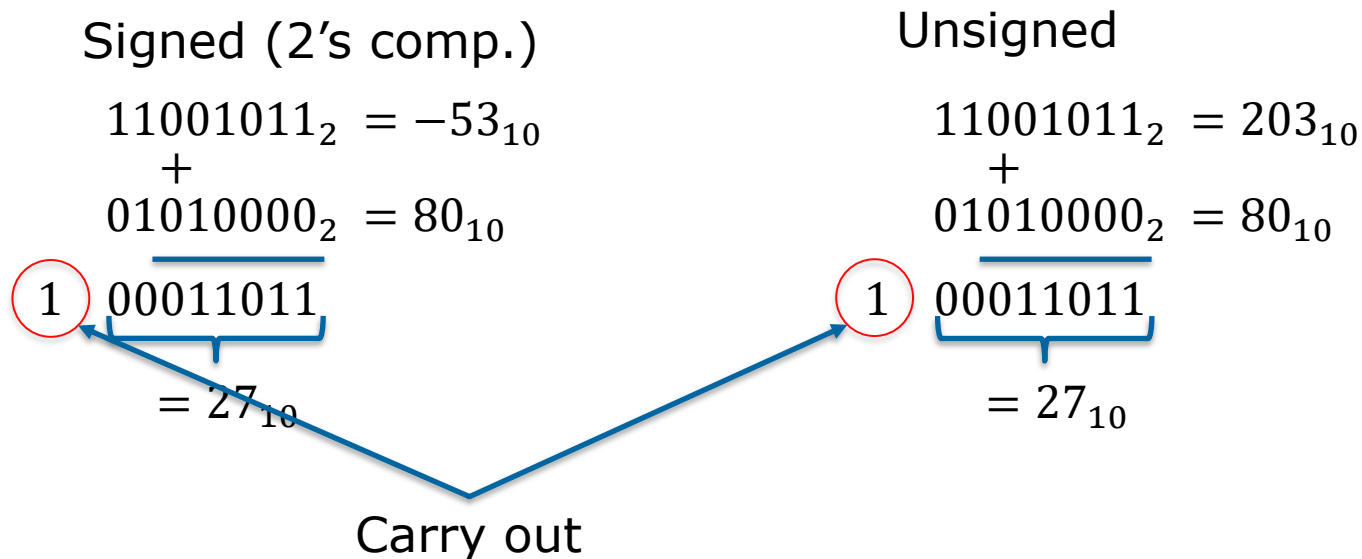
Recap – Number systems

- ▶ We discussed the decimal, binary and hexadecimal systems.
 - $0\text{xCAFE} = 1100\ 1010\ 1111\ 1110_2$
 $= 2^{15} + 2^{14} + 2^{11} + 2^9 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1$
 $= 51966_{10}$
- ▶ We saw how negative numbers are represented in 2's comp.
 - $0b01001111$ represents $2^6 + 2^3 + 2^2 + 2^1 + 2^0 = 79$.
 - It's two's complement (for $N = 8$ bits) is $2^N - 79 = 177$.
 - 177 , as an unsigned integer, has the representation $0b10110001$. $\Rightarrow (-79)$ has a 2's complement representation of $0b10110001$.

$$\begin{array}{rcl} & 0011 & = 3 \\ + & 1001 & = -7 \\ \hline & 1100 & = -4 \end{array}$$

Integers

- ▶ *Signed* n -bit integer can represent $-2^{n-1}, \dots, 2^{n-1} - 1$.
 - E.g., 1-byte (8-bit) integer has the range $-128, \dots, 127$.
- ▶ *Unsigned* n -bit integer can represent $0, \dots, 2^n - 1$.
 - E.g., 1-byte integer has the range $0, \dots, 255$.



Checking overflows in 2's comp.

► Which of the following could overflow?

- (pos)+(neg) } Cannot overflow
- (pos)–(pos) }
- (neg)+(neg) Should be negative
- (pos)+(pos) Should be positive

$$\begin{array}{r} 11001011_2 = -53_{10} \\ + \\ 11010000_2 = -48_{10} \\ \hline 1 \quad 10011011 \\ \hline = -101_{10} \end{array}$$

$$\begin{array}{r} 01001011_2 = 75_{10} \\ + \\ 01010000_2 = 80_{10} \\ \hline 0 \quad 10011011 \\ \hline = -101_{10} \end{array}$$

Extending numbers

- ▶ Consider the 8-bit number 01001011_2 .
 - How do we extend it to 16 bits?
 - 0000000001001011_2 .
- ▶ What about the number 11001011_2 ?
 - If it is unsigned, 0000000011001011_2 .
 - What if it is signed?
- ▶ In 2's complement:
 - $11001011_2 = -2^7 + 2^6 + 2^3 + 2^1 + 2^0 = -53.$
 - $1111111111001011_2 = -2^{15} + 2^{14} + \dots + 2^7 + 2^6 + 2^3 + 2^1 + 2^0 = -53.$

Memory units

- ▶ A *binary digit* (bit) is a single 0/1 digit.
- ▶ Byte is a unit of 8 bits.
 - Trivia: why 8?
- ▶ Kilobyte (KB): $2^{10} = 1024$ bytes.
 - Why not follow the SI prefix Kilo (1000)?
- ▶ Megabyte (MB): $2^{20} = 1024$ KB.
- ▶ Then comes Gigabyte (GB), Terabyte (TB), Petabyte (PB), Exabyte (EB), Zettabyte (ZB), Yottabyte (YB) ...

Working unit size -- *Words*

- ▶ A natural unit of data used by a given processor.
 - We will use MIPS32, but MIPS16 and MIPS64 exist.
- ▶ Typically indicate the sizes of:
 - Registers.
 - Memory address.
 - Instruction size.
 - Floating point variable sizes.
 - ...

Working unit size -- *Words*

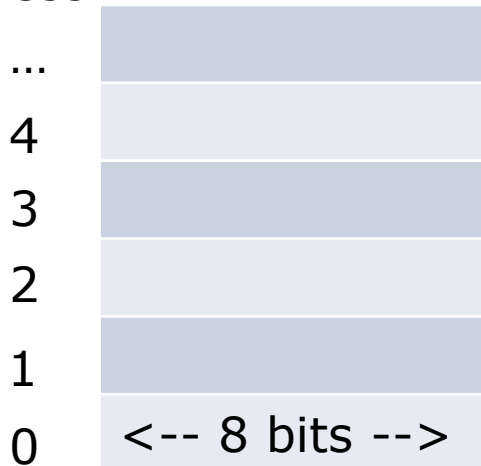
► Terminology:

Size in bits	Name (MIPS32)	Name (Intel x86)
8	Byte	Byte
16	Half word	Word
32	Word	Double word
64	Double word	Quad word

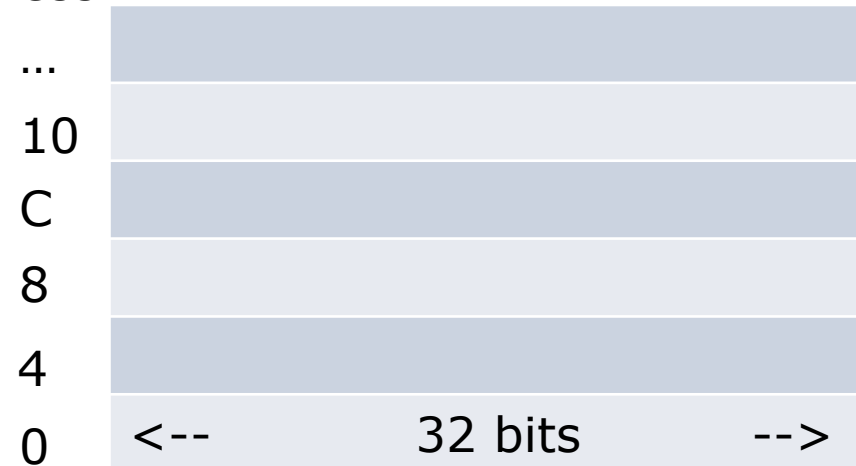
Memory organization

- ▶ An “array” of bytes.
- ▶ Or words.

Address



Address



Byte order

- Consider the word 0xBABACAFE.
 - How'd we store it in address 8?

Offset	0	1	2	3
Address ...				
10				
C				
8	BA	BA	CA	FE
4	0	0	0	9
0				

Big Endian

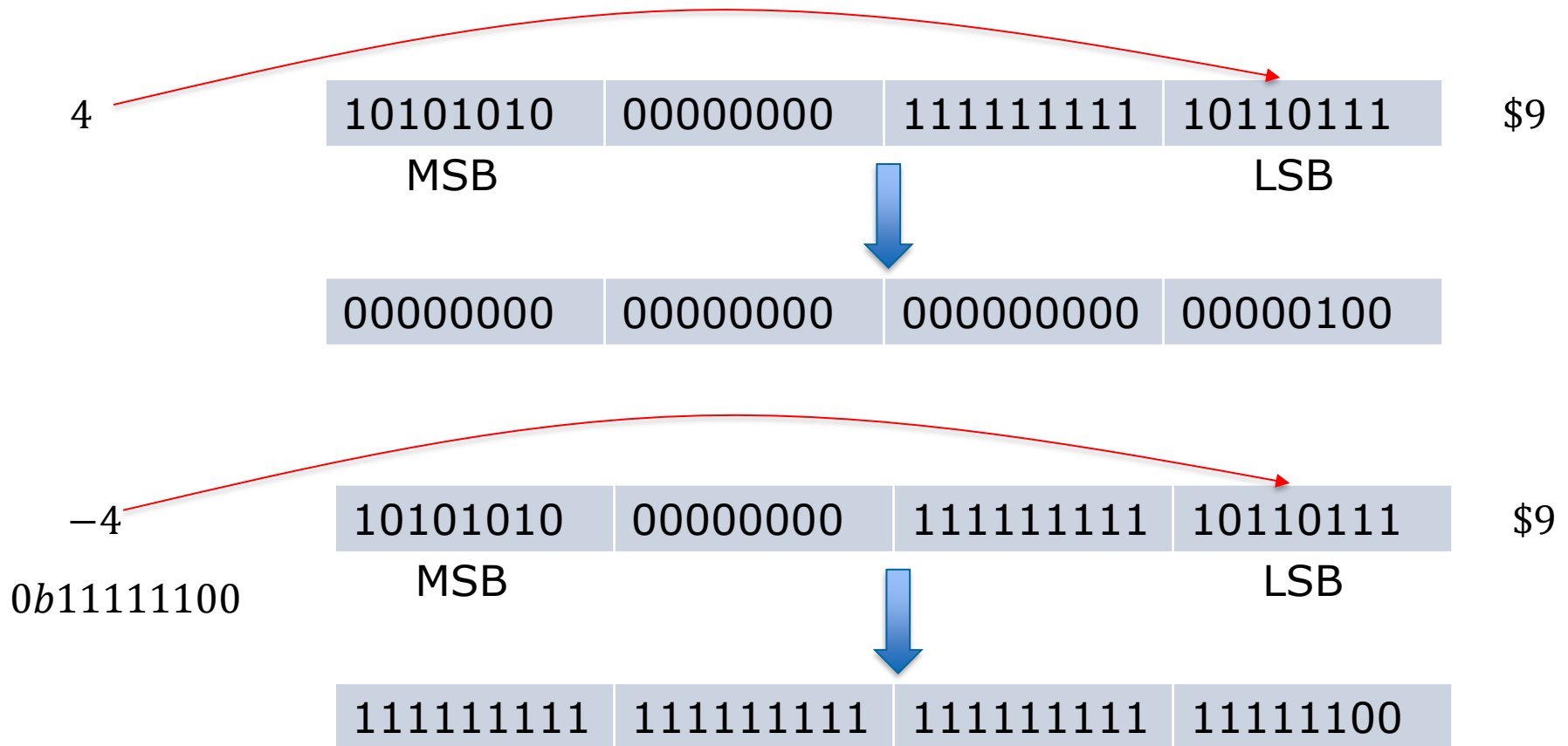
Offset	0	1	2	3
Address ...				
10				
C				
8	FE	CA	BA	BA
4	9	0	0	0
0				

Little Endian

- Say that the 32-bit integer 9 is stored in address 4
 - how'd you cast it to 16 bits in each encoding?

Setting register values

- Suppose we want to write an 8-bit value to (the 32-bit) register \$9.



Summary

- ▶ We discussed the representation range given a number of bits.
- ▶ Saw how to detect overflows in both unsigned and 2's complement arithmetic.
- ▶ Discussed memory units, byte ordering, and what are zero- and sign-extensions.

Pulse Survey

